

ActInf GuestStream 091.1 ~ Dall'Aglio: Consciousness Enjoying Uncertainty

GuestStream | Dall'Aglio: Consciousness Enjoying Uncertainty, Lacan Neuropsychanalysis | 1:43:25

Hello and welcome. This is Active Inference Guest Stream number 91.1 on November 1st, 2024. We're with John Dalaglio et al. discussing Dual Aspect Monism and the Free Energy Principle. We'll have a presentation by John followed by a panel discussion. So thank you all for joining and John for the presentation. To you.

Okay, thanks Daniel and thanks very much to Active Inference Institute for hosting this.

And thank you so much to all of my colleagues and people I very much look up to and such brilliant figures in the field for joining for this. So I'm going to be speaking about the Free Energy Principle and Dual Aspect Monism, specifically from a Lacanian neuropsychanalytic perspective.

And I'll say a little bit about what kind of background I'm bringing from the Lacanian viewpoint.

It's a particular school of psychoanalysis that has mostly engaged less with contemporary natural sciences. And I think that this engagement is very fruitful and introduces some, not necessarily new ideas, but different sort of angles and perspectives. And so I'm really much hoping for a discussion around these ideas.

I'm going to be talking about the Lacanian subject. But to begin with a joke that maybe illustrates some of the ideas. This is actually, this is a joke for the Hegelian subjects, but I think it would equally apply to the Lacanian subject. As the Lacanian subjects or a Hegelian subject, not as something that is a simply straightforward

conscious being, but a certain self contradiction or a certain self loop of the diverting out of one's own self.

And so, so the joke goes that the subject emerges precisely by way of a reflective self-relating reapplication of a logical operator,

as in the worn out joke about the cannibal who ate the last cannibal in the track.

So, and, and these sort of twists show up in a lot of the sort of philosophy behind here, but I think that the ideas

link quite well to some of the computational concepts that I'll be talking about.

So with that, these are some of the sources I'm drawing on. This is a book recently published by the

I'm very grateful for Mark and Carl for such kind endorsements and an article that recently came out in the journal Neuropsychanalysis, discussing some of these ideas and a little more specific philosophical depth.

But to begin, I begin with Solmes's work on dual aspect monism and just to give the background, dual aspect monism is a

philosophical position coming from Spinoza where it doesn't posit some sort of dualism from this Cartesian tradition.

Rather, there's just sort of one stuff in nature. I'm calling it thing in nature, but there's just kind of one thing.

Freud called it the mental apparatus and the, and this is the monism part of it.

And the dual aspects come from the fact that we have two vantage points for this one part of nature.

You can look at this part of nature as a subject, with thoughts, feelings and the like.

And you can look at this part of nature as an object, the brain, the thing that you can poke, squeeze,

and specifically his arguments for what then are the necessary mechanistic features

for a conscious self-organizing system.

So not just sort of things in nature in general, but specifically this part of nature that has this very remarkable quality of being conscious.

And how do we characterize this?

How do we understand it from a free energy principle perspective?

And so the first main point is that within this system, there are multiple categories of homeostatic demand.

And here the main references are innate categories of especially emotional homeostasis, but not only.

Also bodily homeostasis, for example, viable bounds for blood sugar levels for sodium concentrations, but also more socio-emotional things, like it is good for the system to, for the organism to be close to caregivers.

It is good to avoid threats to life and limb in the prehistoric jungle.

When the prehistoric tiger is running at the poor prehistoric human, the one who has, wants to avoid said prehistoric tiger is going to survive.

It's going to maintain the integrity of the system.

And so the argument is that within this system, especially in mammalian brains, there are many, there are many of these categories.

I'm not going to list all of them.

Perhaps we can talk about them later if it becomes relevant, but there are these different systems that are different affective qualities.

I've mentioned a few, say attachment demands, say demands for engaging with novelty, say demands for staying, staying safe from threats to harm.

And computationally speaking, all of these systems function as what's called hyper priors or innate prior predictions.

So within the model, these are predictions that one, the organism is born with, and two, they can't change.

So you can't change your set point for temperature bounds, because if you do, then you dissipate and proteins don't like that.

So, and likewise for the socio emotional homeostats of things, it is good to stay close to caregivers.

These hyper priors do not change.

And so the system has to deal with them.

It has to sort of reckon with the fact of these things.

And therefore, in the very chaotic context of existing, not only in the prehistoric jungle, but also in the complicated interpersonal social life that humans especially find themselves in,

there are many different contexts, and more specifically, many unpredicted contexts.

There's no way that evolution could have predicted that we'd be living in the current kind of society we're living in.

That's not sort of encoded into the gene, into the genome.

But what this means is that there has to be some mechanism within the brain for prioritizing which category am I going to focus on,

or allow to sort of guide my predictive cascades in this given context.

In this given situation, am I going to follow hunger predictive cascades?

Am I going to, I might be hungry right now, but I'm having some sort of thing I want to say, so I'm prioritizing something else.

When you are holding a hot cup of coffee, but you have a goal of wanting to not spill that because you know that might be frustrating you,

then you tolerate the pain that you feel from the hot cup of coffee in your hand.

Rather than following the pain cascade to release the hand and drop it,

you are sort of following a different line of predictions for a moment.

When the child is faced with the conflict of, okay, my loving caregiving caregiver is not doing what I want them to do right now.

That's making me a little angry. I'm actually feeling like I want to destroy and attack this caregiver.

But oh no, that doesn't line up well with my demands to stay close and sort of stay protected by this caregiver.

If I destroy the caregiver, then there's no one for me to be close to.

So I have to choose which one am I going to prioritize, especially in unpredicted contexts.

Within that prioritization is the capacity for choice.

There's the capacity to have voluntary action at this basis.

So these five features are what Mark has argued are the sort of core features of an affectively conscious self-organizing system.

And it's within this model that I want to think about, okay, what does the Lacanian perspective kind of add to this?

And specifically what, how does this open questions for thinking about the free energy principle as we apply it to how we think about especially human ontology?

Within this, we can say, however, that all of these mechanisms, all of these functions that are necessary for there to be affective consciousness,

that sort of subjective quality of feeling of what it's like to be conscious, which is a certain kind of feeling state.

So all of those obey for instance law.

That is, the prioritization function operates in order to minimize free energy.

And how does it do it?

This is the argument that affective consciousness corresponds to precision modulation.

So there are not only within the free energy principle, as again, we all know, predictions and prediction errors.

These are all weighted by precision weights.

Basically the confidence and how confident am I that this prediction will minimize free energy.

Likewise, how salient are these prediction errors?

Should I be attending to them or not?

Can I just dismiss them?

Can I just dismiss them?

And the precision weights are modulated depending on which system, which of the homeostatic systems gets prioritized.

So, for example, and to go back to the poor child who has to deal with conflicts of rage and attachment.

Prioritizing rage adjusts different predictions that are more likely to minimize the,

the prediction errors relating to rage, the frustrations around that.

Attack, destroy, cry, complain, teeth, things like that.

Versus attachment, prioritizing attachment needs would lead to a different modulation.

Different precision weights would be increased.

Others would be decreased.

Others would be decreased that are associated with more affiliative or bonding cascades.

And when it comes to situations of uncertainty, where the organism does not know what to do.

The prioritization here allows one, and this is the argument for why, why affects are conscious or why this precision modulation is conscious is that feeling one's way through one's predictions.

So it's not just that the precision rates change.

That's the sort of what we see thinking from it objectively.

But when we think about it subjectively, this is just feeling.

When you feel your way through your predictions, you're able to feel, okay, this feels better, good in a certain way.

This feels bad, worse in a certain way.

This allows me to sort of feel my way through conditions of uncertainty.

When I don't know what to do and I have to create, I have to explore, I have to invent, I use affect as a guide.

And this is the homeostatic function of affect.

And it is in this way that affect facilitates the minimization of free energy in the long term for the system.

This idea of feeling one's way through one's predictions.

Okay.

So, thinking philosophically here then, what this position argues is that Fristen's Law and the formulas of the free energy principle explain the operations of the mental apparatus, that ontological level, in both of its appearances.

In both subjectively, it explains the qualia and subjective experience of affect, affective consciousness.

And objectively, it explains various neuronal dynamics, neurocascades, specifically neuromodulatory control of postsynaptic gain via these or homeostatic command systems.

And what this allows us to sort of try to begin furnishing, I think this is what's so exciting regarding the philosophy of the free energy principle, is that we can begin to articulate what I'm calling an informatic ontology.

One that is neither physiological nor subjective, but this something else of what it is, it has something to do with information.

It has something to do with minimizing uncertainty and how we go about dealing with uncertainty.

Okay.

What does Lacan have to do with any of this?

And so, Lacan was a French psychoanalyst who is very famous for arguments around language, arguing that one of perhaps his most famous argument is that the unconscious is structured like a language.

And this leads to many arguments and Lacan was notoriously known and Lacanians today are even more notoriously than Lacan, I think known for being remarkably anti-naturalist.

And this is maybe an illustrative quote.

So this is from an interview, I think, I think when Lacan was at MIT or I think somewhere in the States.

We think with our, we think we think with our brains.

Personally, I think with my feet, that's the only way I really come into contact with anything solid.

I do occasionally think with my forehead, but only when I bang into something.

But I've seen enough EEGs to know that there's not the slightest trace of a thought in the brain.

This is not the only comment Lacan makes like this, but it carries a certain sentiment that, and for certain backgrounds for this audience, which I know may not necessarily be Lacanian.

But Lacanian psychoanalysis is very, this perhaps characterizes most Lacanians as sort of not really interested in engaging with the brain that's simply not relevant.

What we care about is language, and that's all we care about.

And it's unfortunate because this really obscures a lot of points Lacan made, and it really misunderstands, I think, how Lacan thought about language.

And I think how Lacan thought about language is actually incredibly consistent with the free energy principle and this idea of information.

So this is a quotation from Lacan commenting on Freud's project for scientific psychology.

And this is a drawing of that Freud has there on the neurons.

And commenting on Freud's use of thinking about neurons and trying to think through this attempt to furnish a psychology for neurologists and so on.

Lacan says, it is this, this idea of neurons, that gives value to atomistic theories.

The latter do not cover any of those things they claim to cover, namely, a certain number of atoms of the neuron apparatus supposedly individualized elements of the nervous system, neurons again.

But on the other hand, the theories of and relations of contiguity and continuity illustrate admirably the signifying structure as such, insofar as it is involved in any linguistic operation.

So what Lacan is saying here is that these individualized elements, the sort of reductionist, one might even say, approach of atomistic theories of the nervous system actually illustrates what he is thinking about through the lens of language and linguistics.

I think this is, this is quite different from saying that it's language per se.

And this is a quote from seminar 20.

I say that the unconscious is structured like a language and I say like, so as not to say, and he says all the time, that the unconscious is structured by language.

The unconscious is structured like the assemblages in question in set theory, which are like letters.

I'm not going to go into Lacan's quasi mathematics here.

But the idea here is that what is the emphasis is on the word like it's not what language is by language, English, German, whatever it might be.

It is what is the nature of language and specifically language as differential system of contiguous and continuous relations.

And Lacan argued, proposed or at least was open to the idea that this can be applied to the brain.

Again, when he's commenting on the project, an orientation to the object, the regulation of the threat, the ideational representatives relate to each other in accordance with the laws of memory organization, a memory complex, a facilitation or a contention.

And I emphasize here whose neurotic apparatus perhaps allows us to glimpse those operations in a material form and whose functioning is governed by the law of the pleasure principle.

Again, and I think this is quite clearly showing how it's not an emphasis on, OK, no, it's only about language.

It's an emphasis on, OK, these linguistic operations, contiguity, continuity can be glimpsed and can appear likewise in the operation of the brain, i.e. the neurotic apparatus here.

And more specifically governed by the law of the pleasure, pleasure principle, which today we can understand as the law of free energy minimization.

And one last quote from Lacan, because I think it's so remarkable here.

This is again in seminar 20, and he's commenting on the on the status of information theory in natural sciences at the time in the early 70s.

And he's talking here that we're at the level of the genes molecular information and the winding of nuclear proteins around strands of DNA that are themselves wrapped around each other, double helix and all of that.

And all of this being tied together and sent via hormonal links.

In other words, messages that are sent, recorded and so on.

Let us note that the success of this formula finds its source in a linguistics that is not imminent, but explicitly formula, not only imminent, but explicitly formula.

In any case, this action extends right to the very foundations of scientific thought being articulated as, and I emphasize negative entropy, a negentropic function.

Again, a resistance of dissipation toward entropy of the system.

So even at the level of DNA, what Lacan is saying is you can see something linguistic here.

Surely there's not speech at the level of DNA in the sense of literal language being exchanged as in what I'm saying right now.

However, it's the logic of communication theory that is relevant here.

And so what we can say then is that when the symbolic or the system of language that is referred to in Lacanian theory, what this refers to is some kind of linguistic organization.

A relation of different elements that are continuous in a chain, ones that relate to each other, ones that are anticipatory, one anticipates the next.

That revolve around a certain antagonism, contradiction or inconsistency.

And this is where the particular Lacanian twist, I think, comes in.

It's that, and this is what Lacan means by the real.

That within any symbolic system, whether that's actual spoken language or the systems of DNA communication or the neuronal atomistic elements within the brain, that there's some inconsistency, some tension that prevents a perfect settling of the system.

It prevents a full sort of resolution where there would be no, let's say, uncertainty, where there would be no antagonism or attention or arousal.

And importantly, this inconsistency is not some wound or imperfection to be remedied or healed or done away with by good enough adaptation, by correction or sort of thing.

It's something that's at the core of one's being.

And Lacan terms this, this neologism estimates, condensation of exterior, intimate, external, internal.

And the idea is that what is most foreign to us, what is most alien to our sense of ourselves is simultaneously what is at the core of ourself.

It is simultaneously what is most precious, what is most essential to how we think, our conscious, self-reflective experience, our, let's say, egoic experience, or even our symbolic sense of where we stand in our organization.

So how can we then conceptualize this idea, this idea of a constituent of a constitutive inconsistency within some sort of communicative or informatic system within this model, within this model that Solms has put forward regarding affective consciousness, free energy minimization, and the like.

First, this requires a crucial philosophical shift.

And this is implicit for some, not for others, but it bears help by stating it explicitly.

And this is what I'm calling following Adrian Johnston, sort of what's called capital N nature.

Capital N nature.

And this idea of capital N nature is an idea that the laws, whatever those laws might be in nature, are fully determinate in that they fully determine the behavior of material particles.

They fully determine what happens by stuff, generally and simply speaking.

And two, they are seamless.

In other words, they're non-conflicting.

They sort of govern in a sort of idealized fashion, all of the operations of whatever there might be.

And then a wonderful exchange between Friston and I forget which journal, it was a philosophical critique of the free energy principle saying that, okay, you're just, you're just saying everything's Markov blanket.

So then you get to use the free energy principle to explain everything.

And I think the response was wonderful.

It was like, yes, it is all Markov blankets all the way down.

And, and one way of potentially reading that and in this preprints that you can find from, from Carl Friston is that this sort of the, the mathematics that follow from delineating things.

Self-organized or self-organizing systems as delineated by the possession of a Markov blanket means that you can actually apply this from the very tiny things of quantum mechanics, all the way up through to the macro scale of classical mechanics.

And the argument here, and this would be an anti-reductionists nightmare is that all of the behavior of these systems from very small to very large can be subsumed within a seamless chain of free energy.

Minimization of obeying something like Friston's law as it gets applied in these different states.

And I am not by any means, an expert on the mathematics here.

So I welcome any corrections on that point, but I think the fear for some would be that, okay, you're saying everything can be reduced to free energy minimization.

And then this leads some people to then say, okay, well, no, there exists something that defies a reduction in this way.

And we can maybe talk about it in a somewhat mystical way as, you know, being or something purely negative or transcendental like that.

Now, the alternative that I want to put forward is that this is not the only way to think about nature.

And this is the perspective from Adrian Johnston, what he calls transcendental materialism and weak nature, so to speak.

And weak nature refers to the idea that it's a materialist philosophical position.

There are only natural material phenomena.

There is no special transcendental subjectivity that only exists because we speak.

There is no sort of special quasi-dualist, but you might, you don't want to say the word dualist sort of a thing that defies capture by any sort of natural science or any sort of material theory.

There are only these phenomena.

However, the laws that govern nature are not seamless.

They are not fully able to assert themselves.

And especially they do not sit in a perfect harmony with each other.

What I'm, what I'm calling weak laws in, in contrast to strong laws that fully determine and line up in a perfect mathematical equation.

Weak laws conflict with each other.

And because of that conflict failed to fully assert themselves.

And therefore, and this is, this is how the sort of transcendental dimension comes in here.

The more than natural, quote unquote, a phenomena that are traditionally described as something transcendental, like human subjectivity.

They emerge not in a deterministic fashion out of the lower levels, but they emerge out of the conflicts and antagonisms of those lower levels.

The points where weak laws come into rift with each other, where they bump up those points are points of non-determinant emergence because they're not emerging from points where determination is working the way it would.

It comes sort of the logic of an accident, the logic of emergence from something that's not working, an emergence from some sort of antagonism.

And a way to kind of put this formulaically complexity breeds negativity, this negativity of what's sort of not positivistically determined.

As things get more complex, there are more opportunities and more risks of things short circuiting and more room for things to go wrong.

And by go wrong, this means something new happens, this sort of emergence by surprise.

And you can think about this, I mean, this is a wonderful book by Daniel Milo, that sometimes we think of evolution as, you know, this strong force of adaptation.

This is a relatively low bar. You can have lots of things inherited.

Think of the number of genetic diseases that get passed down because, you know, and they're not ideal. Those are not adaptive.

Those are not the best for minimizing uncertainty and preserving the state of the system.

Nevertheless, you just need to live long enough to survive and reproduce according to evolutionary law.

It's not a strong law in the sense of producing perfectly adaptive brain or a brain without conflicts that as long as they don't severely undermine this reproductive and survival capacity, they can slip by.

And on the psychoanalytic or Freudian side, this is the arguments and this is not only my argument.

This is or from folks who work in neuropsych analysis, this is arguments from within purely Freudian psychoanalysis as well.

When Freud tries to argue for a beyond the pleasure principle, the sense of a death drive, a drive toward inertia, a drive toward sort of repetition as an innate feature.

In a way, he sort of props up the pleasure principle as then working to do its thing and then, okay, no, when the pleasure principle doesn't, excuse me, work, it's because there's some death drive interfering.

And this is the contrast argument that no, there is no beyond the pleasure principle.

This I'm taking from Jonathan Lear.

There is just the pleasure principle, but the pleasure principle is itself prone to faults, repetitions of prior satisfactions leading it astray and so on.

Simply put, the pleasure principle does its thing, but it doesn't do it perfectly.

It's not a strong law.

The law of the pleasure principle is quote unquote weak law.

So I argue that we can find this kind of logic at this point within the model that Solms has put forward.

So within here, there is the presence of multiple categories of homeostatic demand that have to be contextually prioritized in conditions, especially of uncertainty.

What this is saying is that within the brain, within the level of innate hyper priors within the predictive model, there are conflicts.

There are, there is antagonism.

I gave just one example of the conflict between attachments and rage where you want to remove frustrations to your goals.

These conflicts can easily be multiplied among even just various emotional, socio emotional systems, let alone when we get into the problem of having to think about the fact that other self organizing systems themselves have innate hyper priors and they kind of depend on them to help me with my stuff.

So I need to take that into account.

There's all sorts of opportunities for conflict.

There's all sorts of opportunities for conflict.

And not only that, within this model, these different systems cannot be equated into one common denominator.

There is no sort of equation to say, okay, seven tenths of seeking equals three fourths of rage equals five tenths of fear.

So you can add that up to get, I don't know what the math would be, but whatever, 20th percent of total

overall free energy.

That equation can't happen because these are innate categorically distinct hyper priors.

What you need to do to deal with attachment needs differs from what you need to do to deal with hunger, differs from what you need to do to deal with needs to explore and engage with novelty and uncertainty.

So whichever, and you have to prioritize one because not then you're just sort of helpless before this chaos.

But in prioritizing one, you're necessarily leaving something else uncertain, leaving some residual surprise or or free energy elsewhere, because these cannot be fully subsumed under under one category.

And so it is because of this innate conflict, this innate antagonism, that there is affective consciousness, affective consciousness as the need to feel your way through uncertainty,

to feel your way through the predictions you're employing in.

Okay, I don't know which to prioritize here.

I'm going to prioritize this, but I'm not quite sure what to do.

And then I'd have to kind of feel what feels a little better, what feels a little worse.

This depends on contradiction.

This depends on this innate conflict between categorically distinct domains of free energy, all within a single complex self organizing system.

Now, on the computational side, even though, yes, it is the case that the brain does compute an overall measure of quote unquote total free energy or domain general free energy.

This does not eliminate the difference between these domain specific categories of free energy.

Just because the number is calculated to allow computations and allow action perception cycles to occur, to not sort of just leave everything at a halt, so to speak.

This should not be read as doing away with this difference at the level of the innate informatic ontology.

That calculation facilitates adaptive functioning, facilitates moving through the world, but it doesn't undo the fact that no, there is this conflict.

It's not a conflict that is remedied.

It's one that is a constant sort of tension that persists.

And so, just going back to this sort of diagram here.

What we see, it's not simply that there's just some simply existing thing there.

It's not just some system which does its thing of minimizing free energy.

Well, no, there's actually already there some sort of antagonism or conflict that problematizes a straightforward application of Fristen's law.

It problematizes in the sense that, well, we're in situations all the time where it's not clear what to do, let alone which systems prioritize to perfectly minimize free energy.

And the second let it clinical phenomena testify to this, I think, in a very clear fashion.

So, I just want to close on the main sort of questions I have and sort of ideas I have that I'm hoping we can discuss in addition to anything else that comes up.

But I really want to, I'm curious, and I really think, you know, what are really like the consequences for a self organizing system which contains categorically distinct and contradictory domains of free energy?

What does it mean to think about the free energy principle, where there is this sort of necessary rift within the conscious system of different hyper priors that don't seamlessly fit together?

or two further remarks um you invoke um my uh argument that the different needs of a complex organism um are have to be treated categorically uh they have to be treated as categorical variables not as as it happens because they require you to do different things and those different things conflict with each other but because they each of them have to be met in their own right that's the crucial point uh that it it doesn't help to reduce the total quantity of free energy if it if you leave out one of the categories in other words you always have to breathe as well as drink as well as sleep etc um so it's in this sense that they they they they it's it's it's obligatory that they'd be treated as categorical variables in other words eight out of ten of sleepiness and eight out of ten of thirst are not the same thing they are qualitatively different and incidentally that for me is the ground zero of quality it's why we have to have different qualities in the mind um so you invoke that argument of mine and i'm very uh glad to be to be cited in that way but i just want to make the point

that you don't need to invoke that argument of mine which is as it were built upon or even imposed upon calls um a bigger deeper uh free energy principle i think that it's inherent in carl's own ideas uh for example just to mention one and really the the i could i could think of a few but uh the the inherent antagonism between uh uh complexity and accuracy uh you could have you could have cited that example which is really a pretty basic principle a pretty basic part of the um standard formalism that's used by people working within carl's framework simply put um in in order to maximize accuracy which is a good thing uh you have to increase complexity which is a bad thing uh in order to maximize uh simplicity which is a good thing you have to forfeit accuracy which is a bad thing uh and so there's an inherent antagonism there um and uh as i say i can i could think of several others but that's enough uh to to make the point uh in other words i want to make the point that it's uh it's it's although i'm glad to be cited in this context i think that your argument applies to uh to the the far more fundamental formulation uh that that carl uh has given us then um one last point and remember i'm just trying to get the ball rolling and i'm trying to find is there something we can disagree about um i i'm no philosopher by the way i'm such a badly educated philosopher that i didn't even understand your joke at the beginning about the cannibals uh but um the um i i'm no philosopher but the the the the one thing that's jarred for me uh if i understood you correctly is that you're recommending that a dual aspect monist uh metaphysics be replaced by a transcendental materialist metaphysics uh i can only assume that's what you're saying because i don't see how you can reconcile uh a materialist metaphysics uh with a dual aspect monist metaphysics and i uh because you know the materialist is one side of the dual aspects uh so that's the one thing that jarred for me um i think that uh you know i i cling to a dual aspect monism because uh we want to find a place in um natural science for subjectivity the fact that it exists uh here we are experiencing it right now it's part of nature and we have to find some way uh of of of reconciling it with our with our materialist conception of the universe and uh for me the answer is to uh to recognize that the the material uh perspective is also a perspective in other words there too we're dealing with phenomena uh as observed by sentient minds the subjective and the objective are both perceptual uh vantage points upon something which lies beyond both appearances and that thing which lies beyond both appearances must lie beyond the the the phenomenal properties of matter and uh that's why uh carl's whole way of thinking appeals to me so profoundly you know the the idea i mean it's it's ultimately goes back uh carl's uh way of thinking to galileo telling us that the book of nature is written in the language of mathematics um as i'm speaking i realize um that there's one other thing i might be able to disagree with you about but let me stop there because i i i fear i'm not going to be able to find something uh stronger to disagree with you about than that thank you shall i say something or um so thank you uh thank you mark um uh and um uh perhaps um uh after hearing from some some other folks i'll i'll maybe say more uh less the point of disagreement but more an open question that i have um but at least regarding uh dual aspect monism and transcendental materialism my my my sense is wanting to supplement it uh not to replace and i think this is where there would be a need to re-theorize something in transcendental materialism as it has been uh theorized where um uh i i don't want to say transcendental informaticism or i i'm also not a philosopher um so i'm i'm not i'm

not sure how to say it but basically the to apply that this this way of thinking of uh uh uh uh weak laws
uh emergence from conflict emergence from uh things not working uh out uh that sort of thing to uh to how
we

think about that whatever that thing is in nature beyond um both uh phenomena so but the the way that
i interpret the materialism uh in um transcendental materialism is perhaps closer to um the the the idea
of information and this this is why um like when lacan says uh the the the lacan will say the materiality
of language um uh and this is a materiality not necessarily like the the stuff that they can hold in
everything it's it's a uh it's an open question i think but lacan uh and he changes how he thinks
about that through over time but um in this sense of um the idea of communication the idea of uh uh
information and negative entropy

oh maybe i can uh uh also ask some questions um thank you john first of all uh it's such a pleasure to
hear you kind of tie together all of these different threads um that's uh it's really impressive um
but the first point i'd like to make is maybe um um in relation to adrian johnston's uh um transcendental
materialism like there's in his work also kind of some doubts about how to formulate kind of his own
project i guess um so he has this essay from 2020 uh called uh meta transcendentalism and error first
ontology um where he says that he's come to consider the descriptive label of critical dialectical
naturalism uh synonymous to if not preferable to uh transcendental uh materialism um and then he says
specifically because for him uh it condenses in a single phrase uh kind of idealist methodology and
then realist materialist naturalist uh ontology um so i'd be yeah i'd be curious if there if if
you have reasons to stick or prefer transcendental materialism or whether you would would consider these
um

um um interchangeable um and and my other question um um maybe also as an attempt to i guess stir the
pot in a way in terms of finding a point of disagreement between between you and mark um not not for the
sake

of it but because that's a question i'm quite interested in um um you didn't mention it today but
uh since it's in the title of enjoying uncertainty i'm curious about this point of lacanian drive being
on the side of increases in tension um and in that sense operating outside or beyond the the logic of
homeostasis as as i understand it in your work um and i guess yeah i would i would like to maybe push
back a little bit on that point um to the degree that i wonder whether that kind of resistance towards
the concept of homeostasis doesn't have a lot if not most to do with kind of lacan's concept of
homeostasis uh because correct me if i'm wrong but i think in seminar two he defines it or he says
something like you know homeostasis can mean one of two things the first is uh an equilibrium of a
system and the second thing is dead um and then 10 years later like in the four fundamental concepts
of of psychoanalysis kind of he repeats these same points so i don't think it was it was a quirk of
that time um so i i do wonder if like with the kind of more contemporary concept of of homeostasis
including the seeking system and all of the conflicts that you know arise in that and that you
uh um i think so brilliantly kind of emphasize and and uh uh uh put in the forefront um yeah is it is
is that that rejection or that resistance towards the concept of homeostasis really necessary

yeah thanks ever and i and this would be my uh yeah point for uh kind of question potential disagreement um uh that if we then take the next step it's that at this point uh that i'm carving of affective consciousness as all right arising not from uh free energy minimization but from the contradiction and conflict of these different uh not only categories of granted but even um uh complexity accuracy trade-off um uh within uh the the model that um something else uh comes up there and and uh one way that one way to think about it is that okay yeah there's something here um having to do with not just trying to minimize uncertainty uh uh uh and at least and this is coming into some more of the the the the the the clinical theory um but uh not not only about minimizing uncertainty but finding a way or the fact that um errors and uncertainty is enjoyed in a certain way this is lakhan's idea of um uh that it's an enjoyed not necessarily consciously but uh enjoyed in uh in its own uh uh uh unconscious or it's it's an enjoyment that can't be experienced as such uh the quote-unquote strange satisfaction of the symptom that that fred uh remarks on um but it's this uh i i wonder and this is what i was talking about with um okay what does it mean for a system to try to minimize free energy when it's impossible to to reach that until death um uh as is pointed out um that um yeah there's something about um i think a proactive engagement and enjoyment of uncertainty that i'm not quite sure um is only seeking and this is where i'm i am really curious uh because seeking has its own certain qualitative feel to it uh um the studio uh enjoyments of uh uh uh uh novelty epistemic foraging uh you know the dog running around in the field and you know having fun and whatnot um uh but that that there there's also these points where there's um uh some kind of repetition or some kind of either maintenance or increase of free energy that has a certain uh enjoyment to it um symptomatic enjoyment perhaps and that at least from a lacanian perspective ultimately the goal is not to minimize that free energy uh in the sense of remove the tension uh the goal is not to in to get rid of the symptom or to replace the symptom with a better thing or some better more um adaptive plan uh it's to sort of confront this logic of okay no there is something at the core of one's being that exists because things don't work what's a way what's something that can be done with that uh and this is idea of of of enjoying one's symptom of of uh uh uh and and this is where i i have questions like i don't know what how to think about it necessarily computationally but what would it be to modulate uh something uh uh deep predictions in that sense to modulate them in such a way that uh they repeat so they continue to produce free energy they continue to lead to increases in uncertainty but in a way that uh allows the person to kind of uh find find a vivifying dimension uh to that to their symptom uh to find a vivifying dimension uh uh to how they go about uh managing and uh uh enjoying uh uncertainty rather than uh uh this the at least the illusion it makes to the system just trying to you know totally get rid of it uh in a way um and i think that i think that's where the lakenean position differs i think from from how you've conceptualized it mark where there's it's not uh the idea with um illegitimate predictions prematurely automatized predictions not only the idea of laying down better ones that can hopefully work and be more adaptive but also to to do something different with the prematurely automatized predictions that um sustain ones uh that are sort of your unique way of dealing with the conflicts of these different categories of free

energy um uh and and in that way to do something different with it uh but yeah i'm i'm not sure how that shows up or how that would uh play out uh computational

uh i'm i'm going to resist saying anything more until we've heard from carl

yeah i'm the i'm the eldest so i have to go last really that uh i can't resist slipping in here um

yeah that was a wonderful presentation but actually your last response was was revealing so i learned quite a lot uh in terms of this um you know what mccann brings to the table that you know may have to well i imagine the ambition here is to sort of naturalize that in terms of current thinking about um about the way we understand um sentient behavior and the like um so um before i comment upon the the two sets of questions which i thought were fascinating what i've done and this is really for daniel i've gone through whilst listening to the talk identifying papers that um you may not have reference from other literatures that i think lend weight to your a number of specific points you made during uh the presentation so if there's time at some point close to the end um i've got about four or five um actually six papers that you might enjoy reading completely outside i think your your sort of sphere of reading um that do pick up on some of the sort of fundamentals that you were you were positing as um key key parts of the argument so i'm saying that uh just to forewarn you that if i can at some point share my screen and just flick through because i know this has been recorded then you can go back and

have a look um to see if anything is interesting ranging from girdle's theorem to strange loops you know there's a lot of resonance with um um other formulations outside the other sphere in which we're talking um but just to come back to those sort of key issues um perhaps in reverse order um so this notion of um um enjoying um enjoying the uncertainty um or enjoying um um paradoxical increases in free energy um is there a mathematical formalism that would um resolve that paradox um or sort of paradoxical um account um that would also accommodate um notions of homeostasis as read by lacan um at the same time could be read

in terms of self-organization from a physics perspective or you know certainly from a planful systems perspective

um i two things occur to me first of all your repeated account of um the um going beyond just seeking um as one x manifestation of of this enjoying your your uncertainty um i thought was very interesting because i had assumed when you started answering that this would be exactly the epistemically expected information game that underwrites behavior so when people talk about minimizing free energy that's fine um when one's talking about sense making but when one is talking about how we engage with the world and the decisions and the choices that we make or the things that we select to do

um then one has to think about the expected free energy expected following that choice or decision or act or intention um and um from that perspective that you know the complexity and the accuracy that mark was referring to

now become um risk and ambiguity or when rearranged expected information gain versus um expected cost um and um the expected information gain i think does appear to have a lot of simple explanatory um capacity in talking about the allure of uncertainty and you said that beautifully you talked about

novelty seeking you talked about uh epistemic foraging um and you know i was surprised to hear that that for you

wasn't a sufficient explanation and perhaps it's because you you were associating um the epistemic foraging with seeking of a particular uh of a particular sort but a lot of what you said um um seems to me to be exactly the notion that we enjoy um um pushing the boundaries in the fond hope or knowledge that we will be able to

resolve that uncertainty and become wiser and more um more confident in our engagement with you know with

the world and it it reminds me very much of an argument um put forward for why we actively listen to music um particularly jazz music uh which you know has that exactly the right kind of predictable

unpredictability to keep us engaged and i'm you know unusually the argument here is that um i don't know what's going to happen but i do know i will know if i keep on doing this very much like listening to a joke i don't know what the punchline is going to be but i do know if i commit to this

this uh your direction of travel then i will be amused and someone's you know and that amusement that pleasure will be um if you like scored by the resolution of uncertainty in a volatile and unknown space environment you know um in the space of jokes in that instance um so i i you know i would i and i won't say anything more about that but it'd be interesting if you could discuss what is the the this sort of um

essence of enjoying uncertainty that is um not explained by knowing i'm going to reduce my uncertainty but first i have to actually put myself in a situation where i'm seeking out those uh you know that

potential for resolution of uncertainty about me about mum about the jazz music mathematically um i'm wondering whether one way of sort of bringing lacan into the 21st century would be to go back to

the um something i found quite fascinating which was the two ways of reading homeostasis um as i understand it from whatever it was saying um so um death is certainly one kind of homeostasis um but

clearly it's not the kind of homeostasis that we associate with um biotic self-organization and the mathematical distinction between the two is between equilibrium steady states and non-equilibrium steady states

now all of physics um and certainly the kind of school boy physics you you you would have been exposed to uh when talking about thermodynamic free energy minimization pertains to a point equilibrium so in the language of um dynamical systems there we're talking about a point attractor so at the point you converge to your point attractor everything ceases you're dead you're cold you're a free energy minimum um and you know that's a perfectly apt description of closed systems you know that is what will

happen if you're isolated from a world but when you're open to the world across um your markov blanket for example um then we're talking about a different kind of equal a non-equilibrium uh steady state

and this has a characteristic itinerancy i mean mathematically connect um it's um i can't resist slipping this in because it it also uh appeals again to sort of helmholtzian ideas but in this instance um it has to be the case

in any system that is far from equilibrium or open to an environment that has attained some kind of steady state i.e persists and exists it has to have a solenoidal aspect to it by the helmholtz decomposition

that is a mathematical truism now there are many ways of reading that solenoidal aspect it could be a

biorythm it could be an oscillation it could be um replication dynamics at an evolutionary scale
it could be walking it could be talking could be my heart your everything that returns to some point
in a weak sense so it's not exactly but returns to the neighborhood and i just um i i just pick up on that
because of you know for many reasons but specifically because of your the repetition aspect of of this kind of
um um

um aspect uh sorry of your of that um um the the terminology that you were talking about
well well there is exactly this description of the particular itinerancy that characterizes
um free energy minima that are a result of attaining a non-equilibrium steady state so you have this itinerancy
there can't be any other way of keeping within an attracting set unless you keep returning to the
point that nearby point that you are once that you were once in so i'm just wondering if you want to
go beyond seeking and beyond um epistemic foraging um then perhaps what you're saying is well yes but
this

is in the context that i am on this attracting set that has this quintessentially biotic kind of um
recurrence or or repeatability that gives us an itinerancy among many many different time scales
that makes me the kind of thing that i am and you know in this environment in which in you know to
which to which i'm i'm open so i just put that out there as a possible mathematical metaphor for for
talking about sort of what the camera is referring to if you want a physiological metaphor i think it's
important to actually um distinguish now between homeostasis homeoresis and allostasis and alloresis
if there is such a thing so the distinctions are the homeostasis is fast reflexive whereas the allostasis
actually has some future pointing aspect to it so you can plan to be homeostatic that that in itself is
not homeostasis it you know it brings a certain uh depth and itinerancy and one could argue in virtue
of the depth it takes you much closer to the kind of structures and you know the um reliance upon
um precision modulation that you were talking to but there's another um axis here which is the difference
between stasis and recess so homeoresis is like homeostasis but for paths and of course as soon as
we're talking about um systems that have itinerant orbits where you do have to keep revisiting
characteristic states of being at many many different time scales then the kind of homeostasis we are
talking about is now homeoresis is actually keeping close to that path and i'm just wondering if
Lacan's distinction between the two kinds of homeostasis was really a distinction between
mathematical homeostasis which is a point of tractor and death and homeoresis which is keeping to a path
or a set of paths that define an orbit usually chaotic and it's usually stochastic but it doesn't
have to be i'm talking far too much i'm gonna come i'm gonna go back to the the the um mark's initial
observation just to celebrate that um so you um in my world there's a distinction between
categorical and continuous variables um and i think mark was picking up on that and i think he was
absolutely right to do that i do think that there's a really important role for categorical
generative models um in relation to the description about sort of symbols and language and sort of carving
nature into discrete ways of being i think that's absolutely essential and i'm sure you're right that
some unconscious um processing um you know probably takes place under discrete language like um
um generative models and possibly also um conscious processing um but the i think that's an orthogonal

argument from um the notion the really important notion i thought you were you were trying to get across which is

that there are multiple constraints um that um provide another kind of conflict because i've got sort of multiple dimensions that i have to i have to um comply with it's not just one dimensional pleasure there is pleasure in multiple multiple dimensions and sometimes some of the the excursion along one dimension

may be more um unpleasurable than the excursion along another dimension and that both of those if you like the degree of excursion will be very context sensitive and hence i now have to prioritize do i want to sort of be homeoretic in this dimension or that dimension i think mark was trying to say that it's the multi-dimensional aspect that there are there are more than one discrete dimensions to um the way that we have to

um mathematically in terms of the expected free energy evaluate the cost of this particular plan um and that provides a profound departure from things like reinforcement learning where there's only one um dimension so i've now taken to talking about not expected value or utility but expected cost the negative of that so i can now say that this minimization of expected cost part of the expected free energy um is effectively a multiple constraint satisfaction problem and it's usually if you read the sort of you know the the maths literature in active inference that's usually encoded by c um uh and one of the reasons for c is it stands in for constraints but it's multiple constraints in every dimension that you that there exists an outcome and of course if you now condition c on some inferred state of being or some context um then you have exactly what you were saying which is this problem of actually prioritizing this constraint over the over that constraint the reason i like the word constraint is that um you can now speak to physicists who believe that the free energy principle is dual to james's constrained maximum entropy or caliber principle the end the entropy is for points so with the homeostasis the caliber is for the path so that would be the homeoresis so now you've got a way of talking to physicists because what you're saying is it's all in the constraints and these constraints are multiple there are as many dimensional as as the dimensionality of the phase space so again i'm just trying trying to sort of um join uh enable you to sort of generalize you know that these really important points in a way that might actually have a relatively simple mathematical um basis in fact it's almost unavoidable um and certainly that that notion of multiple constraints and inducing a kind of conflict although you are effectively in a multiple constraint satisfaction resolving that con that conflict by prioritizing exactly as you say those constraints what could also be called prior preferences but um if you think of a constraint or a cost as a negative prior preference or log preference then then i think that will make sense i've got so much more to say but i'm going to stop talking now no thank you cole that's that's really uh fascinating and this helps me i think um uh maybe formulate what i'm trying to think through from the from the lakhan end of things so i i think that the the the the uh i think i think why there would be um some suspicion of the expected free energy calculation as this sort of um uh the enjoying of engaging with novel uncertainty and that sort of thing is because well um generally when it sounds too adaptive um and you know like he knows don't like that but um uh the idea that the the the enjoyment um doesn't derive from the fact that okay i'm going to

engage with this uncertainty because and then i know that i will get that pleasure in the end and then it will meet that need that it's it's um almost enjoyment for its own sake um uh or the increase of uncertainty almost for its own sake but what i think would bring it together is that um it would be uh and my use of the terms might not be correct uh sort of technically but the sort of uh ways of thinking through the expected uh uncertainty would be for uh an anticipated feature that never actually materializes uh an anticipated um outcome and this i think would maybe approximate what what lacan calls the fantasy uh and specifically the unconscious fantasy which is not um it's not uh imagistic it's sort of uh uh it's something that is never uh uh formulated explicitly can only be constructed um uh in the course of a treatment but it's sort of um the virtual terminal points upon which um um various of these scenes of increased repetitions that don't work out um it would be this sort of this increase of uncertainty that leads to bad outcomes um that doesn't ultimately to the um decrease of free energy um and yet it repeats regardless uh and the the the to rationalize that um uh symptomatic let's say uh repetition is by reference to some kind of fantasy formulation of oh you know this is what the pr this is the anticipated outcome that the the person had that uh gives a little sense to why this repetition was happening why this um repeated um uh increase of free energy which is this weird kind of enjoyment um uh without the more as as we typically think about in terms of adaptive uh learning um by evaluating potential costs uh uh for uh different policies and things um and i think that that would be potentially um uh one one way of of bringing those together and it's it's uh uh the homeothesis idea is is very uh interesting where uh there's this diagram i can find it somewhere but there's this diagram lakhan has for the drive um uh and it's uh he draws it as um as as a loop as an arrow loop and circling um negativity and circling something that you never quite hit um and and that's why the drive has this repetitive logic and it repeats uh encircles something that it's aiming at but never quite lands on in part because uh it has a purely uh phantasmatic status that there isn't something that um could finally be hit um and hence uh the intense repetition continues um so perhaps that that would be i think um uh i i think one way to link the uh to link seeking and expected free energy um through expected for energy for a future that doesn't exist uh nevertheless exists in the in the unconscious for the subject um i'll read a question or two from the live chat okay andrew writes where do you place the reality principle in your mapping in seminar seven the con seems to view it as being in service of the pleasure principle so is it a lesser priority concept to incorporate here yeah that that's really big and i think um i don't make a huge distinction because of that that it's still ultimately in the service of the pleasure principle but um one of the points that is also made is that the reality principle not only as a sort of self-correcting uh mechanism in the sense of uh learning and updating uh prior predictions based on uh new experiences in that sense but also lakon talks about reality principle as um uh the index to the the the maintenance of a minimal level of tension in the psyche uh uh the maintenance of um uh from the the the uh the exigencies of life uh that uh cause there to be a certain minimal constant level of tension and that that um that that maintenance uh being linked to what we've been discussing as this okay what what does one then do with this uh uh sort of necessity of repeating uh uncertainty that uh has this uh at least uh subjectively or

phenomenologically this paradoxical uh component to it awesome another question from the chat for you john or anyone

uh kendra wrote does affect consciousness itself emerge as a mistake from the complexity of self-organizing system okay if i can speak metaphorically yes as a mistake but but it's a good mistake and it's in the sense of um uh the um uh slavoj zizek has this uh uh thing of you know it's something that happens by accident but then it was something that turned out to be to be good you know you're trying to you're you're trying to do something something goes wrong then oh wow but this is new and unexpected but it it but it still works in its own way um like the making of um games uh novel invention or the making of uh in history you know different cuisines or different foods or cheeses or wines that uh emerge because something wrong a mistake happened in the fermentation process but oh look now we have this

and now now this is something that's all wonderful uh and i i i think that um that's one nice way to think about clinical work uh uh and from from a lacanian perspective it's that okay can can something differently be done or can something new come from uh this this mistake uh quote unquote mistake of uh how one is uh maintaining a drive circuit or maintaining a certain uh repetition of uncertainty uh that still sustains it but in a different way awesome bucky fuller talked about mistake mystique and it's sort of a living systems perspective with the error first ontology and the vivifying perspective on the differential rather than like dust in a motor which will just inevitably make it more and more off kilter how are you gonna continue with the work or where's the book and everything at uh well the the the book is published very um uh thankfully and um i i i i can't say exactly where where the next step is but um uh going to try and apply this a little more empirically

uh is is is my my personal hope um uh mark i wanted to invite you you said that you had a potential point of uh disagreement and uh uh i just wanted to make sure you could you could air that um i think that given the time constraints uh i won't go down that particular path um but i i'll rather pick up on i i'm also aware that we must leave time for carl to share his screen because i too would love to see those references that he has in mind but um so i'll just make one final remark which has to do with them the main thread of disagreement that i'm pleased to see we did manage to find uh about the the equating of jouissance with uh with seeking um and you said uh almost uh in passing uh and i think almost embarrassingly you said well lacanians don't like uh to uh to to explain through source in this sort of way because it sounds too adaptive um so i i want to comment on that and i saw a a little frown on carl's face when you said that and i shared the frown without expressing it so this this this is what what i think we have to take seriously uh feelings um you know once they evolved they become part of nature and they do something and they actually have causal powers and i think that we all too easily forget that um now uh the the feeling of lust clearly uh the adaptive uh in other words why it was selected in uh why why why it's such a feeling evolved is because it encourages us to perform sexual acts which ultimately result in the uh in reproduction and therefore in the survival of our species uh but that's not what motivates each sentient subject uh to to um to have sex uh what motivates us is the feeling uh and uh the the feeling um very frequently we want the pleasure without

reproducing we we're hoping not to reproduce so the the feeling has motivational power has causal power all of its own quite independently of the uh of the uh adaptive uh mechanisms whereby it evolved now all i wanted to say is the same applies to seeking seeking evolved uh for the very good reasons that we've been discussing uh that it is positively a good thing uh in in terms of uh long-term uh free energy minimization uh to to to positively engage with uncertainty and to increase your your your your free energy in the process but the feeling uh is has as the the same relationship uh to that uh the feeling in other words the the the in my view the the feeling that lacan's talking about the the enjoyment of the increased uh attention uh the the the feeling exists in its own right and and has its own status and uh so you don't know i am engaging with uncertainty now because in the long run this is going to reduce my free energy not at all you're just engaging with uncertainty because it's nice and so it feels good uh you know not why uh and and and and i i i still find that the most satisfactory explanation for the phenomenon that lacan is is is uh trying to explain there thanks i've enjoyed this conversation very much thank you thank you mark ever any last words and then carl you can share your screen

um yes thank you ever i also i thought this discussion was great um yeah i'm also really interested in um this point of how expected free energy may accommodate the phenomena that you described in lacanian terms especially maybe when we're dealing with expected free energy not just in the sense of what will happen based on my current beliefs but maybe especially once you get to the more complicated schemes for example in the case of professor fristen's sophisticated inference where it's not just a matter of what will happen uh based on what i believe but also what will i believe based on what happens please correct me if i if i hope i hope i don't butcher it now but like where the where the whole the whole cascade gets gets um more and more complicated i kind of disconnected from the present as

it as it proceeds um yeah that's that's that's uh what about that

no you didn't a bunch butchers at all that's it it's very sophisticated dimension sophisticated inference in in uh um i i'm wondering whether the point that um mark was alluding to there is that all the ingredients that go into the computation of an expected free energy um and it's just interesting to know that this some of these ingredients uh emerge as things called intrinsic motivation in your robotics so when you're actually um if if you're looking for attachment theory with robots there are people that do this a lot um and they understand that expected the seeking part of a certain kind of an elemental kind in terms of intrinsic motivation which i also find quite quite quite nice um sorry but what i was going to say was um the the the evaluation of all of these moving parts that contribute to the different and potentially conflicting aspects of the composition of the expected free energy rest upon the precision which brings us back to the precision now that doesn't come from magic it has to come from your generative model so there has to be a state of mind that can prescribe and implement and switch on the right precisions in this context in exactly the way i think both john and mark were talking about and i think it's at that level or possibly even a meta level above that the feeling gets into the game um so it's not the is not the intrinsic motivation in and of itself that is felt it is all the enabling of the prioritization via the precisions that will nuance what is intrinsically motivating what you know

what um you know what compels me to seek what compels me to you know and so on so forth and i recognize

that this particular profile of precision is mine and it is um associated with this deployment of the precision and the activation of the mechanisms that would be required to to modulate them so i think the feeling um could be completely divorced from the actual um expected free energy in the sense that you know um even if my expected free energy was was uh non-adaptive in the sense i have absolutely no idea what to do next i'm bored there's no more epistemic value in this environment i have no preferences um you know i just don't know what's going to you know what what to do what to do next i may actually see that as a displeasurable thing or a pleasurable thing um simply because in order to set that um set that profile i've had to induce a particular precision which is apt for this for this particular particular context so i think that you know that metacognitive or deep aspect where um you need to have that particular depth to your to your self-mortals in order to specify what precision is up to for this moment maybe something that that is you know allows you a little bit of latitude in terms of feeling good about what is completely maladaptive and vice versa um i don't know that i i suspect that's that in part was a was a point that uh that mark that mark was making

do you want to share your slides or papers carl and then that will be i do but i i i i can see it's three minutes past um we i could just do this for for john and mark um if um if we're over time just do it just do it all right let's just go and uh um yeah i should apologize i feel a bit naive now i clearly do not understand the repetition compulsion and look at the canyon i have to think more about that so yeah share screen it's gonna share screen in the live chat carl we're trying to understand whether reading your papers is it only about the value of the information

and and the and the uh the expected cost of course um in fact yes so here we were so this is documentary proof that i have now taken to referring to prior preferences as expected cost so that you can use this phrase multiple constraint satisfaction where the cost um um if like scores the constraints and just to re-emphasize it's a multi-dimensional has many dimensions as there are outcomes that one could possibly sense measure infer or or observe um so this is uh from a re the most recent description of the expected free energy um um in biological psychology from the supervised structure learning and that's uh yeah there we are but what i thought was um just coming back to this the the original joke um um about um the cannibal um the cannibal eating um um having eaten the last cannibal on um on the island

i um i i thought that the story was going to be a pact in terms of the um the inability of ourselves to represent ourselves um and um i'm just compelled by chris field's work in this space where he stays close to the physics in his own words to to demonstrate that um self-representation at some level of recursion and some level of depth is a mathematical impossibility so i just wondered whether whether that was um um what was being intimated uh you know by that joke it transpired that it probably wasn't so important in the central argument but uh i just put this um paper up um uh by chris fields um and mike levin uh just as a recent treatment of this from the point of view of the physics um there is also a treatment of this from the point of view of consciousness um um where in a slightly

more narrative form this you know this impossibility of seeing oneself or representing oneself at a certain depth or recursion or sophistication in an economic sense um inherits from the notion of an irreducible markov blanket that can't be subdivided on the inside so the only way that this particular irreducible markov blanket or system in the brain which could be distributed and would certainly include um the um the sources of neuromodulation in the spirit of uh mark's thesis which is explicitly mentioned in this paper um the only way that it can see itself is by acting upon the rest of the brain for example that action can only be through deploying precision um so i thought there was a nice point of contact between the purely mathematical treatment of self-representation and how it might be unpacked this is a paper which has um gone through many very versions but um um uh talks about the inner screen model of consciousness which is on archives first also author uh ramstead um but there chris was a key uh key um author of um um on that um the um that was just reiterating oh yes um i talking about sort of um weak laws um

um um could be read in many different ways um and um you know what one aspect of this um could be that we're not talking about deterministic systems and we're talking about stochastic systems that can only be described probabilistically another aspect of this is um the departure from deterministic systems as we move

into real um or um certainly from the physics perspective biotic systems that have um a uh a chaotic aspect that is often called uh or um uh has a mathematical characteristic of strangeness which is um um exploited in this treatment of strange particles um with a sort of rather uh uh with poetic license also a reference to um the strange loop of hofstadter um but in this instance speaking exactly to this recurrence so i now appreciate this is not quite the repetition compulsion it's not fundamental but the recurrence that we were talking about before in terms of revisiting states that one uh once was in um but also it's unpacked in a particular way in terms of what the physics brings to the table in terms of different kinds of things um um and noticing that um it is certainly possible to represent your infer your own actions and that may be the first step towards a very deep model towards this irreducible um markov blanket uh where we can no longer represent ourselves but certainly can um at subsequent or lower levels have a mechanism for understanding the feelings of the um the um the remainder um so this is a paper by me um with strange things in the title um this oh dear this site yeah um this one was interesting um i'd read this with care but it it does um provide you uh an entree into um to into a literature um which actually in this instance comes from uh fintech or financial uh from um formal theories in um in finance and and the like um but it has two aspects which i thought were interesting first of all um it it sort of reiterates the the focus that at the end of the day we are quantum um or we can be read as quantum systems so that the things have a language like property in the in the in the sense that they have a certain discreteness um and secondly the implications for that um taking a slightly different view from chris fields when it comes to representing self and important aspects of representing self are not just in terms of self-modeling and the um ensuing mechanisms for feelings but in the immune system and being able to recognize that this cell this kind of this antigen um belongs to me or it belongs to a virus is a really important um

aspect of self representation self-recognition so i just put this paper up by sherry marcos who is um fascinating writer but i repeat it this is difficult stuff but it's it does highlight biotic elements have unique digital identifiers with gene codes executing 3d self-assembly for morphology and so on um and her argument uses the recursive operation of self-reference um by appeal to um um you can read that godel's formulation of uh of sentences uh and the implications of self-replication uh representation the interesting thing is she takes the argument um from maths applies them to the thymic self in the spirit of immunology but then takes it into the realm of mirror neurons which is getting slightly closer to uh um don't leave that one yeah um just a couple of notes just to say that um you know the early work um speaking to the importance of neuromodulation and precision has a lot of empirical

backing in terms of certainly dopamine encoding the precision of beliefs about what i'm going to do next that i have this categorical aspect so there's a lot of work um suggesting that you know things certainly things like dopamine have a profound role um in the feelings associated with things like seeking this is the work of um a group of people who are committed to understanding phonology in a sort of philosophical sense that highlights and i'm sure you know about this but really does highlight um the importance of precision and specifically conditioning precision on higher level representations of self here referenced in terms of hidden mental states that themselves are um updated in accord with fairly more or

or less precise beliefs whose precision themselves are um encoded by a higher level

uh um in the sense of the sense that highlights and i'm sure you know about this but really does highlight um the importance of precision and specifically conditioning precision

um um is known to make this is known to make this is known to make this known to make this known to make this

in accord with fairly more or less precise beliefs whose precision themselves are encoded by a high level so this is one example of this contextualizing by being able to recognize yourself in certain states of mind either sub-personally or not that enables that are associated with the deployment of precision at lower levels and so on all the way up until you reach this irreducible Markov

blanket so this is work high level states condition the likelihood mapping precision at the level below

so I thought this was a if you did want to sort of formalize this and start to simulate these things

this is the kind of model I think you'd refer to and Lars Sandford-Smith is a sort of one of the key authors in in that space so towards a computational technology of mental action and then finally oh

yeah we don't need to worry about that but I just put that up to remind myself that more recent

leveraging of these discrete states based models now can be read purely in terms of

in your inductive inference of a logical symbolic sort which if you like is another way of

motivating the importance of this language like way in which we build beliefs about what we are going to

do but that can be read at one level when they're very precise as intentional behavior so I thought that

might be an interesting endorsement of the some of the points that were being made earlier on and I

would stop my sharing that was it

thank you all wonderful thank you so much thank you all right good luck with everything John thank you all for joining

bye bye

uh yeah I understand um uh folks have to go can I just at least for recording's sake um uh if folks have to go that's fine just so I can say and this will help me remind myself what I'm saying yeah um people can depart yeah people can depart the live stream continues and then

John if you want to add any closing notes yeah just so on the adaptive points um uh and I I think that a lot of the resistance is just resistance uh even clinically speaking um uh but um I think um the the the the the trick and the challenge and I I think the

uh the the the papers you referenced um uh and shared I I I haven't read them but um I I from how you described them

I think that they uh would be approaching similar uh ideas um but the the difficulty of uh again addressing this point that sort of resists full sort of capture or resists full sort of uh uh resolution or explanation and and how exactly to think about uh that especially when uh as as I said one will kind of repeat something

in and it doesn't lead to uh and it doesn't lead to uh an adaptive outcome and yet this continues to repeat but then what to do with it in a way that allows uh someone to kind of uh enjoy a bit a bit differently or or in in another way um that I think would be fine to be called um adaptive but again uh uh adaptive maybe in in in in this sense of uh maintaining these sort of different paths of um again error uh again uh paths of um

uh expected free energy that uh could continue to sort of persist um uh yeah so just to kind of uh tie that together I I think that that's um uh the the the cannibal who makes the mistake of eating the last cannibal and then okay what what what is he to do now um uh with with his oral drive um so I'll I'll stop with that but thank you again so much um this was really good for you guys to be able to do this

thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you thank you thank you thank you thank you thank you thank you thank you thank you thank you
thank you

Thank you.